PPP-S-1927A
December 10, 1976
SUPERSEDING
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FEDERAL SPECIFICATION

SACK, SHIPPING, PAPER (REINFORCED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for two types of glass fiber reinforced shipping sacks where water resistance, puncture and tear resistance is required. The sacks may be used for exterior or interior packaging of items weighing not more than 10 pounds.

1.2 Classification.

1.2.1 Types. Sacks shall be furnished in the following types, as specified (see 6.2):

Type I - Plain kraft paper.
Type II - Extensible kraft paper.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-268 - Paper, Kraft, Wrapping.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-97 - Tape; Pressure-Sensitive, Filament Reinforced.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Tapes, Packaging, Adhesive and Gummed; Methods of Inspection, Sampling, and Testing.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

Technical Association of the Pulp and Paper Industry (TAPPI) Publications:

T 402 - Conditioning Paper and Paperboard for Testing.
T 404 - Tensile Breaking Strength of Paper and Paperboard.
T 410 - Basis Weight of Paper and Paperboard.
T 433 - Water Resistance of Paper and Paperboard.
T 457 - Stretch of Paper and Paperboard.
T 475 - Bleeding Resistance of Asphalted Paper at Elevated Temperature.
T 803 - Puncture and Stiffness Test of Container Board.
T 414 - Internal Tearing Resistance of Paper.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Liners.

3.1.1.1 Type I. The kraft paper used to fabricate the sacks shall conform to UU-P-268, grade B, as specified in table I.

3.1.1.2 Type II. The paper used to fabricate the sacks shall be a nominal 30 pound, plus 5 percent basis weight (24 x 36-500) unbleached kraft, highly extensible in both directions. The minimum elongation at tensile failure shall be 4.75 percent machine direction and 3.25 percent cross direction; when subjected to the test specified in 4.4.5.

3.1.2 The sacks shall conform to the requirements specified in table I. The strength properties shall be determined in accordance with 4.4.5.

TABLE I.

Nominal basis weight (24 by 36-500) ±5%	Minimum average beach puncture	Minimum average tear strength		Minimum average tensile strength per inch	
		Machine direction	Total machine direction plus cross direction	Cross direction	Total cross direction plus machine direction
Type I 100 lb.	32 Beach units (inch-ounce) inch	716 grams	1,750 grams	35 lb.	105 lb.
Type II 100 lb.	27 Beach units (inch-ounce) inch	900 grams	2,000 grams	18 lb.	52 lb.

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3.1.3 Liner laminants. The laminated liners shall meet the following requirements.

3.1.3.1 Water resistance. The liners shall have a water resistance sufficient to prevent a definite color change when tested for not less than two hours in accordance with 4.4.2.

3.1.3.2 Bleeding. The sacks shall not show any evidence of bleeding or staining when tested as described in 4.4.4.

3.1.3.3 Adhesives and laminants. Adhesives and laminants used in the fabrication of the sacks shall adhere and seal the exterior and interior sheets at all edges. The seams and bottom fold or flap shall not burst when subjected to the test specified in 4.4.3.

3.2 Construction.

3.2.1 General requirements. Sacks may be of the conventional flat or gusseted type with a back seam flap of 1-inch minimum. The bottom flap or fold shall form a seam of 1 to 1-1/2 inches. All seams, folds or overlaps shall be formed with water resistant adhesive conforming to requirements of 3.1.3.3. The sack shall be scored 1-inch from the open end. The material may be serrated at the open end of the sack.

3.2.1.1 Additional requirements. The walls of the sacks shall consist of 2 plies of kraft paper combined with water resistant laminant and a nonwoven glass-fiber reinforcement. The reinforcement shall be lengthwise fibers spaced 3/4 to 1-inch apart, with widthwise fibers perpendicular to the lengthwise fibers; or, the fibers may be arranged in a diamond shaped pattern with 3/4 to 1-inch spacing between the parallel sides of the diamond.

3.3 Dimensions. Sack dimensions shall be outside measurements, range from 4 to 20 inches in width and 8 to 40 inches in length, exclusive of flap. The dimensions shall be expressed as width by length. Overall dimensions, as specified, shall be accurate to within 1/4 inch (see table IV and 6.2).

3.4 Workmanship. Workmanship shall be first class throughout. Bottom and center seams of the sacks shall be securely sealed. The sacks shall be free from defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified, a lot shall consist of all sacks of one dimension submitted for delivery at one time.

4.3 Sampling.

4.3.1 For visual and dimensional inspection. For each lot, a random sample of sacks shall be selected in accordance with MIL-STD-105, for visual and dimensional inspection. The inspection level shall be S-2 and the AQL in accordance with 4.4.1.

4.3.2 For tests. A random sample of 40 sacks for testing shall be selected from each lot. Ten sacks shall be used for each of the tests in 4.4.2, 4.4.3, 4.4.4 and 4.4.5. Failure of more than one sack in any one test shall be grounds for rejection of the lot.

4.4 Examination. Samples of sacks selected in accordance with 4.3.1 shall be examined for conformity with the requirements, packing, and marking provisions of this specification.

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4.4.1 Classification of defects. Inspection of sacks shall be conducted as follows:

Major defects, AQL 1.0 percent defective.
1. Material not as specified (see 3.1).
2. Construction not as specified (see 3.2).
3. Dimensions not within tolerances (see 3.3).

Minor defects, AQL 2.5 percent defective.
1. Exterior or interior of sacks torn (see 3.4).
2. Seams not sealed (see 3.2).

4.4.2 Water resistance test. A six inch square piece of sack wall, not including a fold, seam or overlap, shall be cut from each sample sack. Each sample shall then be tested for water resistance in accordance with TAPPI T-433, except that the stipulations concerning the number of tests to perform and the reporting of average results shall not apply.

4.4.3 Seam strength test. The sample sacks shall be filled to 3/4 of the total length above the bottom of the sack with dry sand or any other free-flowing material with a density greater than 1-gram per cubic centimeter. The weight of sand or free flowing material added shall not exceed ten pounds. The sacks shall then be subjected to five drops from a height of 36 inches, or until failure occurs. The drops shall consist of two drops on the flat sides, one drop on the long edge, one drop on the short edge and one drop on the corner of the sack. In all cases, the strength of the back seam and bottom seam or fold shall exceed the burst strength of the bag (see 3.1.3.3).

4.4.4 Bleeding and staining test. The sample sacks shall be tested for bleeding or staining in accordance with TAPPI T-475, Bleeding Resistance of Asphalted Paper (see 3.1.3.2).

4.4.5 Strength tests. Each sample (table II) shall be tested in accordance with the following current methods:

Test	TAPPI No.
Atmospheric Conditions	T 402
Basis Weight	T 410
Puncture Test	T 803
Tensile	T 404
Extensibility	T 457
Tear strength	T 414

4.4.6 Examination of preparation for delivery. An examination shall be made to determine that packing and marking as required by section 5 are complied with. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an Acceptable Quality Level (AQL) of 2.5, expressed in terms of defects percent defective.

TABLE II. Examination of preparation for delivery

Examine	Defect
Marking	Missing; incorrect; illegible; of improper size; location, sequence, or method of application.
Materials	Components missing, damaged, or otherwise defective.
Contents	Number of components per container is more or less than required.

4.4.6.1 Examination of palletization. An examination shall be made to determine whether palletization as required by section 5 is complied with. Defects shall be as specified in table IV. The sample unit shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5, expressed in terms of defects percent defective.

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TABLE III. Examination of palletization

Examine	Defect
Finished dimension	Length or width exceeds specified maximum requirements.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps specified.
Weight	Exceeds maximum load.
Marking	Omitted; incorrect; illegible; or improper size, location, sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The sacks shall be packed in a box conforming to PPP-B-636, class weather-resistant, grade V2s. Unless otherwise specified, the quantity per container shall be as specified in table V. The box shall be closed and strapped in accordance with the appendix to the box specification. In lieu of steel strapping, boxes shall be reinforced with tape conforming to PPP-T-97, type IV, applied in accordance with the appendix thereto.

5.1.2 Level B. The sacks shall be packed in a box conforming to PPP-B-636, class domestic. Unless otherwise specified, the quantity per shipping container shall be as specified in table IV. The box shall be closed in accordance with the appendix to the box specification.

5.1.2.1 When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.1.3 Level C. The sacks shall be packed to insure carrier acceptance and safe delivery at destination at lowest rates, in containers complying with the Uniform Freight Classification or National Motor Freight Classification as applicable.

TABLE IV.

Dimension	No. bags per shipper
4 x 8	2,000
6 x 10	1,000
8-3/4 x 12	1,000
9-1/2 x 14-1/2	500
10-1/2 x 16	500
12-1/2 x 19	250
14-1/4 x 20	250

5.2 Palletization (military requirements). When specified (see 6.2), sacks, packed as specified, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3 Marking.

5.3.1 Military shipments. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special markings required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

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6. NOTES

6.1 Intended use.

6.1.1 Reinforced sacks are frequently used for interior and exterior packaging in the shipment of publications and small parts where water resistance is required. In some cases, it may be desirable to add cushioning materials inside the sack to provide protection against breakage.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, dimension and quantity required.
(c) Level of packing required (see 5.1).
(d) When palletization is not required (see 5.2).

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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